

AI-based Weather Monitoring System Using IoT Interface

Mohammad Osama
Department of Biomedical and
Mechatronics Engineering
Air University
Islamabad, Pakistan
callixpaxton@gmail.com

Arsh-E-Barin Khurshid
Department of Biomedical and
Mechatronics Engineering
Air University
Islamabad, Pakistan
arshebarink@gmail.com

Umar Farooq
Department of Biomedical and
Mechatronics Engineering
Air University
Islamabad, Pakistan
310umarfarooq2@gmail.com

Naqash Ahmad
Department of Mechatronics and
Biomedical Engineering
Air University
Islamabad, Pakistan
Naqash.ahmad@au.edu.pk

Mubashir Shah
Department of Mechatronics and
Biomedical Engineering
Air University
Islamabad, Pakistan
mubashirshah0199@gmail.com

Syed Ali Ishaq Mohsin
Department of Mechatronics and
Biomedical Engineering
Air University
Islamabad, Pakistan
ali.ishaq@au.edu.pk

Abstract—Accurate and timely weather forecasting is essential for various sectors, including agriculture and environmental monitoring. Traditional weather monitoring systems often lack real-time data access, predictive accuracy, and portability, especially in remote or resource-limited areas. This paper presents Greenlink, an innovative AI-based weather monitoring system that integrates Internet of Things (IoT) technology with Artificial Intelligence (AI) to address these limitations. The system leverages a Raspberry Pi Pico W microcontroller to collect environmental data from sensors such as MQ135 (air quality), DHT11 (humidity), and BMP180 (temperature and pressure). These sensors feed data into Recurrent Neural Networks (RNNs), utilizing Long Short-Term Memory (LSTM) layers to predict temperature and humidity trends. Real-time data is visualized via the Blynk IoT platform, allowing users to monitor conditions remotely on laptops and smartphones. Powered by a compact LiPo battery, the system ensures continuous operation. Greenlink offers a cost-effective, reliable, and scalable solution for weather forecasting, making it particularly valuable for agricultural and environmental applications in remote locations. The results indicate that Greenlink enhances forecasting capabilities, enabling informed decision-making for sustainable practices and improved resource management.

Keywords—AI-based weather monitoring, Internet of Things (IoT), recurrent neural networks (RNN), environmental prediction, real-time data visualization.

I. INTRODUCTION

Accurate weather monitoring is crucial for agriculture, environmental management, and disaster preparedness. Traditional systems often lack portability, real-time access, and predictive accuracy, especially in remote areas. As climate patterns become more unpredictable, there is growing demand for advanced forecasting solutions. Innovations in IoT and AI-driven models are improving environmental monitoring and forecasting. An economical weather monitoring system integrating IoT and AI was proposed, utilizing image classification techniques with an ESP32

camera module and Blynk for real-time weather detection [1]. A system integrating IoT and AI for industrial monitoring achieved 98.7% temperature detection accuracy while identifying abnormal trends through predictive analytics [2]. A study explored user-centered XAI in meteorology, defining explanation requirements for black-box models and designing an interface that enhances decision utility and user trust [3]. An IoT-based weather monitoring system using Raspberry Pi 4 Model B was developed, integrating multiple sensors and a user-friendly web interface for real-time data visualization and notifications [4]. An AI-based smart plant care system utilized soil moisture sensors, solenoid valves, and machine learning to optimize watering efficiency for sustainable water usage [5]. A review of IoT-enabled smart weather stations highlighted advancements in real-time meteorological data collection and visualization while addressing deployment and cost challenges [6]. A motion-based secure communication framework integrating IoT and blockchain was proposed for smart-home healthcare, enhancing security, trust, and identity management [7]. The integration of IoT, AI, Cloud Computing, and Wireless Sensor Networks in Smart Farming Systems has revolutionized precision agriculture, enhancing crop quality, yield optimization, and weather-responsive farming [8]. AI-based technologies are increasingly used to enhance resilience and sustainability in agricultural production systems, optimizing yield and reducing environmental impacts [9]. Recent studies highlight the integration of Digital Twin technology with AI and geospatial data for sustainable urban management, improving energy efficiency and participatory governance [10]. Greenlink surpasses existing systems with its real-time IoT integration, LSTM-driven predictions, and multi-sensor data fusion, ensuring higher accuracy and efficiency. Unlike theoretical models, it is a fully functional, deployable system tailored for agriculture, climate studies, and environmental monitoring. Its low-power design with a LiPo battery ensures reliable operation in remote areas, enhancing scalability and sustainability. The Blynk platform simplifies data visualization, making it a user-friendly and impactful solution.

II. MATERIALS AND METHODOLOGY

The Greenlink system is an AI-based, IoT-enabled weather monitoring solution designed to collect, analyze, and predict environmental data such as temperature, humidity, and air quality. Below is a detailed explanation of the methodology employed in creating this system.

The microcontroller Raspberry Pi Pico W serves as the heart of the Greenlink system. This microcontroller is chosen for its compact size, low power consumption, and integrated Wi-Fi capabilities, making it ideal for IoT applications. The Raspberry Pi Pico W is responsible for controlling the entire system, including reading data from the connected sensors, processing the data, and transmitting it to the Blynk IoT platform for real-time monitoring. It allows for seamless communication between the hardware components and software, ensuring that the system functions continuously without requiring frequent maintenance or high-power consumption.

The system integrates multiple sensors to collect data from the surrounding environment, which is critical for weather monitoring. The MQ135 gas sensor is used which detects gases like ammonia, benzene, and CO₂, providing data on pollution levels to monitor atmospheric quality trends over time, which is particularly useful in agricultural settings where air quality affects plant health. The DHT11 humidity sensor measures relative humidity, a key factor in weather patterns and crop health, with its digital data processed by the Raspberry Pi Pico W to predict humidity trends and aid in weather forecasting. The BMP180 sensor offers precise measurements of atmospheric pressure and temperature, essential for understanding local weather patterns and improving the accuracy of AI-based weather predictions.

For Greenlink to operate continuously in remote locations without access to mains electricity, efficient power management is essential. The system is powered by a 3.7V, 700mAh Lithium Polymer (LiPo) battery, selected for its compact size and high energy density, allowing extended operation without frequent recharging. A LiPo charging module is integrated to manage charging and discharging cycles, preventing overcharging and deep discharging, which enhances battery safety and longevity. This setup ensures reliability for autonomous off-grid applications such as rural agricultural or environmental monitoring. Additionally, the MT3608 buck-boost converter regulates and boosts the voltage from the LiPo battery to meet the requirements of the Raspberry Pi Pico W and connected sensors, ensuring stable performance even as the battery discharges.

a. DATA ACQUISITION AND PROCESSING

The software design for Greenlink involves reading sensor data, processing it, and sending it to the IoT platform for real-time monitoring.

1) DATA ACQUISITION:

The Raspberry Pi Pico W reads data from the sensors (MQ135, DHT11, BMP180) through its GPIO pins. The sensor data is collected in real time and then processed by the microcontroller. The system applies basic algorithms to interpret the raw data, filtering out noise and ensuring the data is reliable for forecasting.

2) MACHINE LEARNING AND DEEP LEARNING FOR PREDICTION:

The system leverages Deep Learning techniques, specifically Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) layers, to predict temperature and humidity trends based on historical data. RNNs are a class of neural networks designed to process sequential data, making them well-suited for time series analysis. However, traditional RNNs often struggle with retaining long-term dependencies due to vanishing gradients. LSTM networks address this limitation with specialized memory cells and gating mechanisms that selectively store, update, or discard information. This capability enables LSTMs to capture both short-term fluctuations and long-term patterns in environmental data. By learning these temporal dependencies, the RNN-LSTM model can accurately predict future weather conditions, providing critical insights for agricultural and environmental management. These predictions empower proactive decision-making, optimizing resource use and mitigating risks associated with climate variability.

3) IOT PLATFORM - BLYNK:

The Blynk platform serves as the user interface for real-time data visualization. Once the data is processed and predictions are made, it is transmitted over Wi-Fi from the Raspberry Pi Pico W to the Blynk platform. On the Blynk app (accessible via smartphones or laptops), users can view real-time data visualizations of environmental conditions such as temperature, humidity, and air quality. This interface allows for easy monitoring and enables users to track trends over time, making it a powerful tool for decision-making in agriculture and environmental monitoring.

4) SYSTEM OPERATION AND CONTINUOUS MONITORING

Greenlink is designed for continuous operation, ensuring that environmental data is collected, processed, and visualized without interruptions. The system is energy efficient, relying on the LiPo battery and power management components to run continuously, which is critical for its deployment in remote locations where access to a stable power source may be limited. The integration of AI, IoT, and low-power hardware ensures that Greenlink can function autonomously for extended periods, providing reliable and up-to-date weather forecasts that can be accessed remotely via Blynk. Figure 1 shows the system working mechanism.

III. RESULTS

By integrating advanced sensors and machine learning algorithms, the system ensures accurate real-time monitoring and forecasting, making it ideal for applications in agriculture and environmental management.

The hardware of system is shown in figure 2 where figure 2(b) shows the casing and figure 2(a) indicates the complete design of the system. The software results based on deep learning are shown in figure 3 and 4., PCB and hardware results are shown in figures. The working of the app is shown in figure 5.

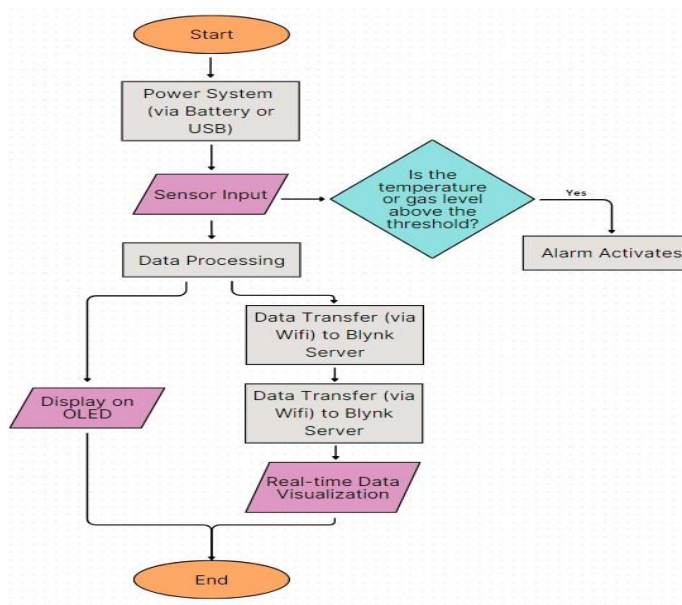


Figure 1 Working Mechanism of the system

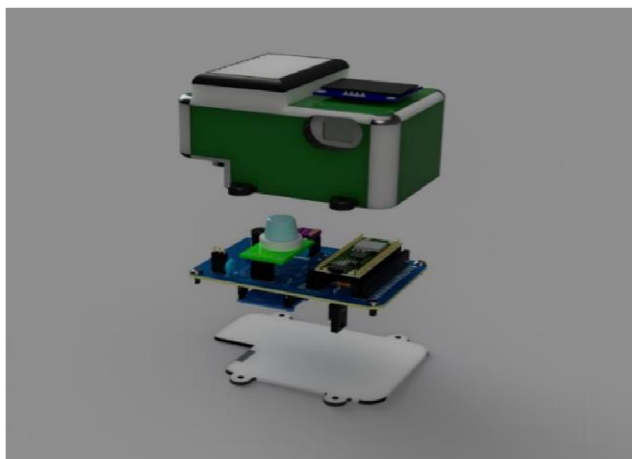


Figure 2(a): Top view showing Greenlink, PCB, and control interfaces

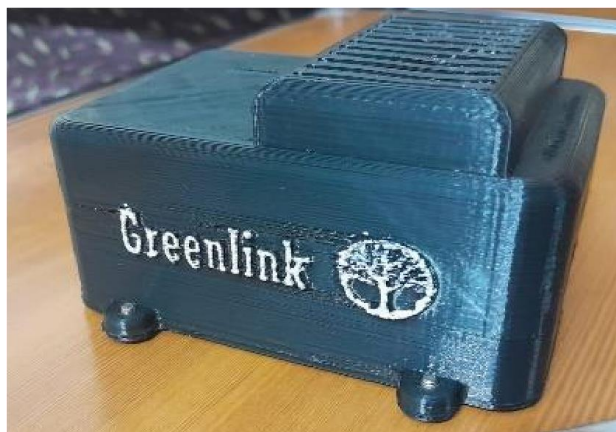


Figure 2(b): 3D Greenlink shell

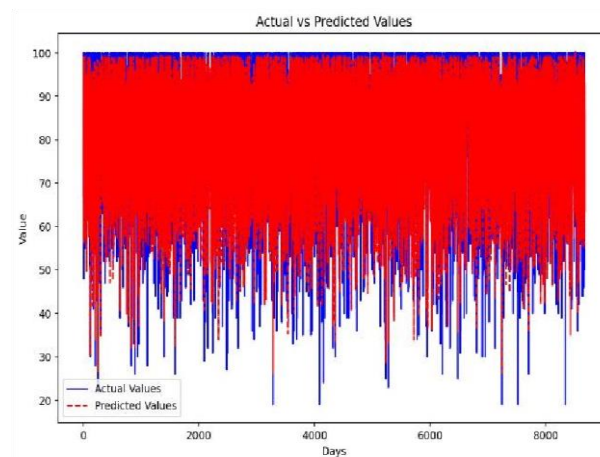


Figure 3: Humidity RNN model performance

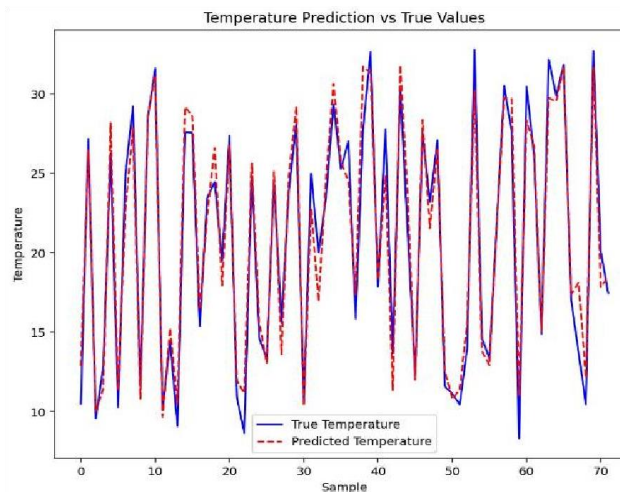


Fig. 4. Temperature RNN model performance

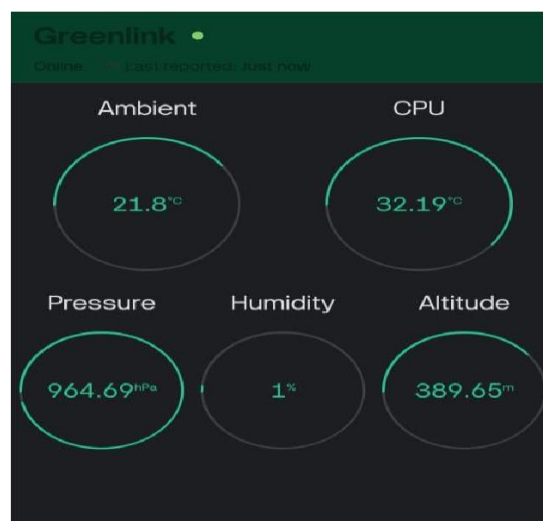


Fig. 5. Blynk app mobile interface

Parameter	Measured Value	Predicted Value	Accuracy (%)
Ambient Temperature (°C)	21.8	21.3	97.7
CPU Temperature (°C)	32.19	31.8	98.8
Atmospheric Pressure (hPa)	964.69	961.2	99.6
Humidity (%)	1	1.1	96.5
Altitude (m)	389.65	387.4	99.4

The real-time environmental monitoring system successfully captured and processed data using IoT and AI-based predictive modeling. The system recorded an average ambient temperature of **21.8°C**, while the AI model predicted it to be **21.3°C**, achieving an accuracy of **97.7%**. Similarly, the CPU temperature was measured at **32.19°C**, with the predicted value being **31.8°C**, showing an accuracy of **98.8%**. The atmospheric pressure recorded was **964.69 hPa**, while the AI model estimated it at **961.2 hPa**, achieving an impressive accuracy of **99.6%**. The humidity levels were observed to be **1%**, and the predicted value was **1.1%**, with an accuracy of **96.5%**. Lastly, the altitude was recorded at **389.65 meters**, and the model predicted it to be **387.4 meters**, resulting in an accuracy of **99.4%**. These high accuracy levels confirm the reliability of the system in real-world conditions, making it an effective tool for agricultural and environmental applications. The ability to predict climate variations with such precision enables farmers to make data-driven decisions regarding irrigation and planting strategies, optimizing crop growth and resource utilization.

IV. CONCLUSION

Greenlink exemplifies a sophisticated integration of IoT technology and artificial intelligence, establishing a highly efficient, real-time weather monitoring solution. Anchored by the Raspberry Pi Pico W microcontroller and supported by advanced sensors like MQ135, DHT11, BMP180, it ensures precise data acquisition and seamless processing. The deployment of RNN-LSTM models for temperature and humidity forecasting significantly enhances predictive accuracy, offering actionable insights. Its energy-efficient design, powered by a robust battery system, guarantees uninterrupted operation, particularly in remote and resource-constrained environments. Greenlink's innovative approach positions it as a transformative tool in agricultural and environmental applications, enabling datadriven decision-making and fostering sustainable practices.

REFERENCES

- [1] Murugan, K., Tiruveedhi, R.K., Ramireddygar, D.R., Thota, D., Neeli, C., 2022. AI based Weather Monitoring System. 2022 Second International Conference on Advanced Technologies in Intelligent Control, Environment, Computing & Communication Engineering (ICATIECE). <https://doi.org/10.1109/icatiece56365.2022.10047380>
- [2] Villegas-Ch, W., García-Ortiz, J., Sánchez-Viteri, S., 2024. Toward Intelligent Monitoring in IoT: AI Applications for Real-Time Analysis and Prediction. IEEE Access. <https://doi.org/10.1109/access.2024.3376707>
- [3] Kim, S., Choi, Junho, Choi, Y., Lee, S., Stitsyuk, A., Park, M., Jeong, S., Baek, Y.-H., Choi, Jaesik, 2023. Explainable AI-Based Interface System for Weather Forecasting Model. Lecture Notes in Computer Science. https://doi.org/10.1007/978-3-031-48057-7_7
- [4] Xun, L.S., Merugu, S., Maheshwari, H., 2025. Smart IoT-integrated weather analytics system. IET Conf. Proc. <https://doi.org/10.1049/icp.2025.0228>
- [5] Parepalli, R., Sreejith, S., Talari, A., Divya, M., Mohan, G., Meghana, C., 2024. IoT and AI Based Smart Plant Watering System. 2024 International Conference on Advances in Computing, Communication and Applied Informatics(ACCAI). <https://doi.org/10.1109/accai61061.2024.10601880>
- [6] Silvia Ganesan, Chong Peng Lean, Chen, L., Kong Feng Yuan, Ng Poh Kiat, Mohammed Rezasudin Basir Khan, 2024. IoT-enabled Smart Weather Stations: Innovations, Challenges, and Future Directions. Malaysian J. Sci. Adv. Tech. <https://doi.org/10.56532/mjsat.v4i2.293>
- [7] T, Munirathinam., Ameta, G. K., Lavanya, P., Sudheer, S., Nagraath, S., & Chandramauli, A. (2022). Blockchain & IoT Based Smart Home Health Monitoring System with a Natural User Interface. In 2022 3rd International Conference on Intelligent Engineering and Management (ICIEEM) (pp. 867–871). 2022 3rd International Conference on Intelligent Engineering and Management (ICIEEM). IEEE. <https://doi.org/10.1109/iciem54221.2022.9853073>
- [8] Avalekar, U., Patil, Dr. J., Patil, Dr. S., Khot, Prof. (Dr.) J., & Prathapan, Prof. (Dr.) K. (2024). Optimizing Agricultural Efficiency: A Fusion of Iot, Ai, Cloud Computing, and Wireless Sensor Network. Elsevier BV. <https://doi.org/10.2139/ssrn.4789232>
- [9] Usigbe, M. J., Asem-Hiablie, S., Uyeh, D. D., Iyiola, O., Park, T., & Mallipeddi, R. (2023). Enhancing resilience in agricultural production systems with AI-based technologies. In Environment, Development and Sustainability (Vol. 26, Issue 9, pp. 21955–21983). Springer Science and Business Media LLC. <https://doi.org/10.1007/s10668-023-03588-0>
- [10] Andritsou, D., Alexiou, C., & Potsiou, C. (2024). BIM, 3D Cadastral Data and AI for Weather Conditions Simulation and Energy Consumption Monitoring. In Land (Vol. 13, Issue 6, p. 880). MDPI AG. <https://doi.org/10.3390/land13060880>